Temozolomide with Radiotherapy (RT) and Adjuvant Therapy-Patients greater than 65 years

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status*
Adult patients greater than 65 years with newly-diagnosed glioblastoma multiforme not suitable for the standard radiotherapy regimen in combination with temozolomide.[i]	C71	00461a	CDS

* This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.

Temozolomide is administered orally once daily with concomitant radiotherapy for 21 days from day 1.
Four weeks after completion of the temozolomide and radiotherapy concomitant phase:
Temozolomide is administered orally once daily on days 1-5 of a 28 day cycle for up to 6 cycles of monotherapy treatment or until disease progression or unacceptable toxicity develops.

Temozolomide and Concomitant Radiotherapy (RT) Phase			
Drug	**Dose**	**Route**	**Cycle (21 days)**
Temozolomide	75mg/m^2 ONCE daily concomitantly with radiotherapy	PO[a]	Continuously with radiotherapy for 21 days
PJP prophylaxis **(Refer to local policy)** is required when temozolomide is administered with concomitant radiotherapy			

4 WEEK BREAK FOLLOWED BY:

Temozolomide Monotherapy Phase (Adjuvant)				
Day	**Drug**	**Dose**	**Route**	**Cycle (28 days)**
1-5	Temozolomide	150mg/m^2 ONCE daily	PO	1
1-5	Temozolomide	[b]200mg/m^2 ONCE daily	PO	2-6
[a]Temozolomide is available as 5 mg, 20 mg, 100 mg, 140 mg, 180 mg, or 250 mg hard capsules				
[b] If patients tolerate cycle 1 at 150mg, starting dose should be increased to 200mg/m^2 for cycle 2-6				
Temozolomide hard capsules should be administered in the fasting state. The capsules must be swallowed whole with a glass of water and must not be opened or chewed. If vomiting occurs after the dose is administered, a second dose should not be administered that day				
If a dose is missed, the patient should make up that dose, unless the next dose is due within 12 hours.				

ELIGIBILITY:

- Indications as above
- ECOG 0-2
- Adequate renal and hepatic function

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EXCLUSIONS:

- Patients with hypersensitivity to temozolomide or any of its listed excipients
- Hypersensitivity to dacarbazine
- Severe myelosuppression
- Significant hepatic dysfunction
- Pregnancy or breastfeeding

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist

TESTS:

Baseline tests:

- FBC, renal and liver profile
- Blood glucose
- Virology screen -Hepatitis B* (HBsAg, HBcoreAb)

*(Reference Regimen Specific Complications for information on Hepatitis B reactivation)

Regular tests:

Temozolomide and Concomitant Radiotherapy (RT)

- FBC weekly up until and including 4 weeks after last fraction of radiotherapy (i.e. Week, 1,2,3,4,5,6,7)
- Renal and liver profile 1 week and 4 weeks after last fraction of radiotherapy

Temozolomide Monotherapy Phase (Adjuvant)

- FBC, renal and liver profile every 28 days (at day 1 of each cycle)*

(* Excluding cycle 1 - the "4 weeks after the last fraction of radiotherapy" evaluation to count towards "day 1 of cycle 1" as well.)

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

Temozolomide and Concomitant Radiotherapy (RT)

- Any dose modification should be discussed with a Consultant
- No dose reductions are recommended, but delay or discontinuation of temozolomide administration should be decided weekly according to haematological and non-haematological toxicity criteria (Table 1 and Table 2 below).
- If temozolomide is interrupted due to adverse events, radiotherapy should continue. When treatment with temozolomide resumes, daily doses should be taken until the last day of radiotherapy as per the original schedule, i.e. missed doses of temozolomide will not be replaced or restored.

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Table 1: Temozolomide dosing interruption or discontinuation for haematological toxicity during concomitant radiotherapy and temozolomide

ANC (x 10^9/L)		Platelets (x 10^9/L)	Dose of Temozolomide concurrently with RT
≥ 1.5	and	≥ 100	75mg/m^2
≥ 0.5 and <1.5	and/or	≥ 25 and <100	Hold until recovery then resume daily dosing at 75mg/m^2
<0.5	and/or	<25	Stop

Table 2: Temozolomide dosing interruption or discontinuation for non- haematological toxicity during concomitant radiotherapy and temozolomide

Adverse Event	Grade	Dose of Temozolomide Given Concurrently with Radiotherapy
Nausea or vomiting	3	Hold until recovery to ≤Grade 2 and then resume daily treatment at 75mg/m^2
	4	Stop
All other non-haematological adverse events (except alopecia)	2, 3	Hold until recovery to ≤Grade 1 and then resume daily treatment at 75mg/m^2
	4	Stop

Temozolomide Monotherapy Phase

- Any dose modification should be discussed with a Consultant.
- Dose in Cycle 1 is 150 mg/m^2 once daily for 5 days followed by 23 days without treatment.
- At the start of Cycle 2, the dose is escalated to 200 mg/m^2 if the CTC non-haematological toxicity for Cycle 1 is Grade ≤ 2 (except for nausea and vomiting (≤3), absolute neutrophil count (ANC) is ≥ 1.5 x 10^9/L, and the platelet count is ≥ 100 x 10^9/L.
- If the dose was not escalated at Cycle 2, escalation should not be done in subsequent cycles.
- Once escalated, the dose remains at 200 mg/m^2 per day for the first 5 days of each subsequent cycle except if toxicity occurs.

Table 3: Temozolomide dose levels for monotherapy treatment

Dose Level	Temzolomide Dose (mg/m^2)	Remarks
-1	100	Reduction for prior toxicity
0	150	Dose during Cycle 1
1	200	Dose during Cycles 2-12 in absence of toxicity

Table 4: Temozolomide dose reduction or discontinuation for haematological toxicity during monotherapy treatment

ANC (x 10^9/L)		Platelets (x 10^9/L)	Dose of Temozolomide concurrently with RT
≥ 1.5	and	≥ 100	Treat on time, no dose modification
<1.5	and/or	<100	Delay start of cycle by 1 week intervals, to a maximum of 3 weeks[a], until recovery, then treat with same dose as day 1 of previous cycle
<0.5	and/or	<25	Stop
[a]If the counts have not recovered after 3 weeks then adjuvant temozolomide should be stopped			

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Table 5: Temozolomide dosing interruption or discontinuation for non- haematological toxicity during monotherapy treatment

Adverse Event	Grade	Dose of Temozolomide Given Concurrently with Radiotherapy
Nausea or vomiting	3	Delay start of cycle by 1 week intervals, to a maximum of 3 weeks [b] until recovery to ≤Grade 2 and then treat with same dose as day 1 of previous cycle
	4	Stop
All other non-haematological adverse events (except alopecia)	2, 3	Delay start of cycle by 1 week intervals, to a maximum of 3 weeks [b] until recovery to ≤Grade 1 and then treat with same dose as day 1 of previous cycle
	4	Stop
[b]If not recovered after 3 weeks then adjuvant temozolomide should be stopped		

Renal and Hepatic Impairment:

Table 6: Dose modification of temozolomide in renal and hepatic impairment

Renal Impairment		Hepatic Impairment	
CrCl (mL/minutes)	Dose	Impairment level:	Dose
≥ 36	No dose adjustment is needed	Child-Pugh A/B:	No dose adjustment is needed
< 36	No need for dose adjustment is expected	Child-Pugh C:	No need for dose adjustment is expected
Haemodialysis	No need for dose adjustment expected		
Recommendations for drug Temozolomide from Giraud et al 2023			

SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic AntiCancer Therapy (SACT) Induced Nausea and Vomiting Available on the NCCP website

Temozolomide: Moderate to high (**Refer to local policy).**

For information:

Within NCIS regimens, antiemetics have been standardised by the Medical Oncologists and Haemato-oncologists and information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS: None

OTHER SUPPORTIVE CARE:

- PJP prophylaxis during concomitant RT phase **(Refer to local policy)**
- Temozolomide can have genotoxic effects. Therefore, men being treated with it should be advised not to father a child up to 6 months after receiving the last dose and to seek advice on

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cryoconservation of sperm prior to treatment, because of the possibility of irreversible infertility due to therapy with temozolomide.

ADVERSE EFFECTS:

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details.

REGIMEN SPECIFIC COMPLICATIONS:

- **Hepatitis B Reactivation:** Patients should be tested for both HBsAg and HBcoreAb as per local policy. If either test is positive, such patients should be treated with anti-viral therapy. **(Refer to local infectious disease policy**). These patients should be considered for assessment by hepatology.

DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for more information.

REFERENCES:

1. Perry JR, Laperriere N et al. Short-Course Radiation plus Temozolomide in Elderly Patients with Glioblastoma. NEJM 2017;7;376:1027-37
2. Giraud E L, Lijster B D, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment: an update. Available at: https://pubmed.ncbi.nlm.nih.gov/37269847/
3. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf
4. Temozolomide (Temodal®) Summary of Product Characteristics. Last updated: 30/10/2019 Accessed March 2025. Available at: https://www.ema.europa.eu/en/documents/product-information/temodal-epar-product-information_en.pdf

Version	Date	Amendment	Approved By
1	22/02/2018		Prof Maccon Keane
2	22/04/2020	Reviewed. Update of emetogenic potential and adverse events	Prof Maccon Keane
3	13/06/2025	Reviewed. Number of cycles updated to 6. Treatment dose modification. Updated exclusions section. Renal and Hepatic dose modifications updated to align with Giraud et al 2023. Regimen updated in line with NCCP Standardisation.	Prof Maccon Keane

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

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[i]This regimen is outside its licensed indication in Ireland. Patients should be informed of the unlicensed nature of this indication and consented to treatment in line with the hospital's policy on the use of unlicensed medication and unlicensed or "off label" indications. Prescribers should be aware of their responsibility in communicating any relevant information to the patient and also in ensuring that the unlicensed or "off label" indication has been acknowledged by the hospital's Drugs and Therapeutics Committee, or equivalent, in line with hospital policy.

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